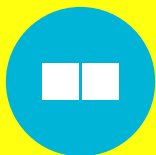
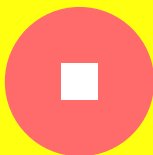


Data Validation & Conditional Formatting

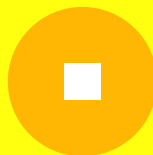
A visual guide to smarter, cleaner spreadsheets



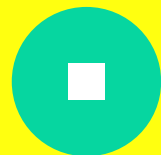
Data
Validation



Conditional
Formatting



Rules &
Criteria



Error
Handling

Automate · Validate · Visualise

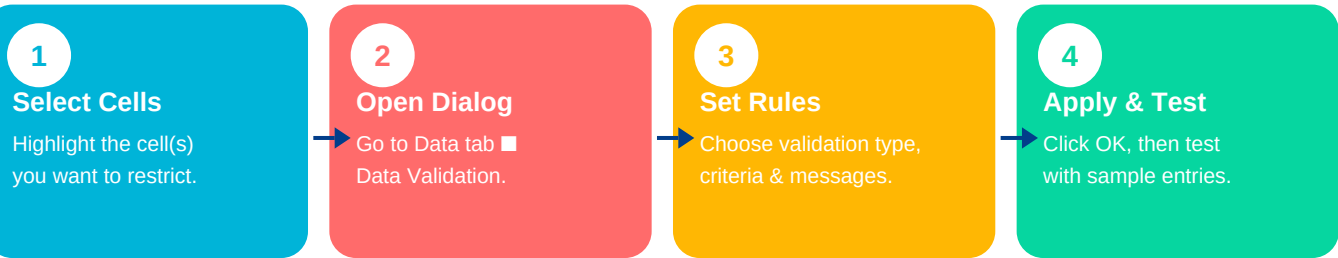
Data Validation

Control what users can enter into your cells

What is Data Validation?

Data Validation is an Excel feature that restricts the type or range of values a user can enter in a cell. It prevents errors, enforces consistency, and guides data entry with messages and dropdown lists.

How to Apply Data Validation



Validation Types at a Glance

Type	Use Case	Example
Whole Number	Numeric ranges	Between 1 and 100
Decimal	Floats & ratios	Greater than 0.0
List	Dropdown menus	Yes, No, Maybe
Date	Date entry control	After 01-Jan-2024
Time	Time range control	Between 09:00–17:00
Text Length	Limit string length	Less than 50 chars
Custom	Formula-based rules	=AND(A1>0,A1<1000)

Messages & Alerts

Input Message

Shows a tooltip when cell is selected. Guides users before they type.

Select cell → Data Validation → Input Message tab

Error Alert

Stops, warns, or informs when invalid data is entered. 3 style options.

Stop  | Warning  | Information 

PRO TIP: Use Custom validation with formulas like =ISNUMBER(A1) for powerful, flexible rules. Combine with Named Ranges in list validation so dropdowns update automatically when source data changes.

Conditional Formatting

Automatically highlight cells based on their values

What is Conditional Formatting?

Conditional Formatting applies visual styles — colors, icons, data bars — to cells that meet specified criteria. It lets you instantly spot trends, outliers, duplicates, and patterns without writing a single formula.

Rule Types



Highlight Cell Rules

Greater than, Less than, Between, Equal to, Text Contains, Dates



Top / Bottom Rules

Top 10 items / %, Bottom 10 items / %, Above / Below Average



Data Bars

Horizontal bars inside cells, proportional to cell value



Color Scales

2 or 3-color gradient across a range; Min → Mid → Max



Icon Sets

Arrows, traffic lights, star ratings, flags based on thresholds



New / Custom Rule

Formula-based rules; e.g. =MOD(ROW(),2)=0 for banded rows

Applying Conditional Formatting – Step by Step

1

Select Range

2

Home > CF

3

Choose Rule

Pick a preset or 'New Rule' for a formula.

4

Set Format

Choose fill color, font, border style.

5

Manage Rules

Use 'Manage Rules' to edit priority & scope.

Month	Sales	Bar	Status
Jan	42		Low
Feb	78		Mid
Mar	55		Mid
Apr	91		High
May	33		Low
Jun	67		Mid

Advanced Techniques

Power-user tips & best practices

Data Validation vs Conditional Formatting

■ Data Validation

- Controls INPUT
- Prevents bad data entry
- Shows dropdown lists
- Triggers error alerts
- Applied before entry
- Found in: Data tab

■ Conditional Formatting

- Controls APPEARANCE
- Highlights existing data
- Colors, icons, data bars
- Visual cues only
- Applied to existing values
- Found in: Home tab

VS

Power Tips

■ Named Ranges

Use named ranges as list sources in DV so dropdowns update automatically.

■ Dependent Dropdowns

Use INDIRECT() to create cascading dropdown (Country → City).

■ Formula-based CF

=WEEKDAY(A1,2)>5 highlights weekends automatically in a schedule.

■ CF with DV

Combine both: restrict input AND colour-code by status simultaneously.

■ Manage Rule Order

CF rules evaluated top-down; use 'Stop If True' to prevent overlap.

■ Clear Rules Safely

Use Manage Rules dialog — don't use Clear All which removes all rules.

Keyboard Shortcuts

Alt + A + V + V

Open Data Validation dialog

Alt + H + L

Open Conditional Formatting menu

Ctrl + Shift + L

Toggle AutoFilter (often paired with DV)

F5 → Special

Select cells with Data Validation rules

Ctrl + `

Toggle Show Formulas (review CF formulas)

7+

Validation Types

6+

CF Rule Categories

100%

Formula Compatible

∞

Custom Rule Possibilities

Quick Reference Cheatsheet

Data Validation + Conditional Formatting

DATA VALIDATION

Allow: Whole Number

Data → Data Validation → Allow: Whole Number

Dropdown List

Allow: List → Source: A1:A10 or type items

Custom Formula

Allow: Custom → =AND(A1>0,A1<=100)

Input Message

Data Validation → Input Message tab

Error Alert – Stop

Style: Stop → rejects invalid entry

Error Alert – Warning

Style: Warning → allows after confirm

Circle Invalid Data

Data → Data Validation → Circle Invalid

Clear Validation

Data → Data Validation → Clear All

Dependent Dropdown

Allow: List → Source: =INDIRECT(B1)

Whole column validation

Select entire column, then apply DV

CONDITIONAL FORMATTING

Highlight > value

Home → CF → Highlight Cells → Greater Than

Top 10 items

Home → CF → Top/Bottom → Top 10 Items

Data Bars

Home → CF → Data Bars → choose style

3-Color Scale

Home → CF → Color Scales → choose preset

Traffic Light Icons

Home → CF → Icon Sets → 3 Traffic Lights

Formula Rule

New Rule → Use a formula → =A1>AVERAGE(\$A\$1:\$A\$10)

Highlight duplicates

Home → CF → Highlight → Duplicate Values

Highlight weekends

Formula: =WEEKDAY(A1,2)>5

Banded rows via CF

Formula: =MOD(ROW(),2)=0

Manage/Delete Rules

Home → CF → Manage Rules → Edit/Delete

Remember: Data Validation = Control Input | Conditional Formatting = Visualise Output

Use them together for bulletproof, beautiful spreadsheets.